

# 370M people search Google Photos each month, and each one finds the photo only if four steps all work

1.5B+

people use Google Photos each month

9T+

photos and videos stored

370M+

people search their photos each month (1 in 4)

150M

Google One subscribers paying for storage

**A photo you can't find is a memory you've lost**  
**Goal:** more users retrieve a photo they remember but can't precisely describe. **Why Google cares:** a library people can't search is worth less to keep, and storage is what Google One sells.



**It becomes one number: User Retrieval Rate (URR)**  
Of the people who search for a photo they can only vaguely describe, the share who reach it within 28 days. It counts people, not searches.

$$URR = E \times [1 - (1 - I \times S \times R)^n]$$

**E Express it**

Can they put the memory into words at all?  
Today: a search box, or a question to Ask Photos

**I Be understood**

Does Photos read the clue the way they meant it?  
Today: matches people, places and things

**S See it surface**

Is the photo in the results at all?  
Today: a grid of matching photos

**R Recognise it**

Can they tell it is the right one?  
Today: thumbnails with date and place

**n Try again**

How many tries before they give up?  
Today: retype the search

**How to read it:** a person succeeds only if they can express the memory and one try passes I, S and R together. One weak step caps the whole number, so the next slides find which step breaks.

# 85,140 public posts became 144 comparable failed searches, showing what was kept, what was lost and where search broke

1 · COLLECT

85,140

Public posts  
(Play Store,  
App Store, YT,  
Reddit)

2 · SCREEN

819

Relevant (Gate  
A 5,305 → Gate  
B 1,333  
sample)

3 · EXTRACT

144

Specific  
attempts (720  
episodes; 62  
scoreable)

4 · AUDIT

203

Pairs checked  
blind by Qwen  
27B model

5 · COMPARE

9 Areas

Ranked only on  
agreed fields  
(O1–O9)

| Extracted Field in Schema             | The Brief's Underlying Question Answered                | Reliability Check   |
|---------------------------------------|---------------------------------------------------------|---------------------|
| asset_type                            | What kinds of old photos do users struggle to retrieve? | Validated in Gate B |
| cues_retained (14 cue types)          | What do people actually remember about the photo?       | Jaccard 0.557       |
| cues_lost (date, place, album, words) | What have they forgotten when search fails?             | Jaccard 0.673       |
| query_verbatim · search_mode          | How do they search when memory is incomplete?           | Quote verify 85.2%  |
| failure_stage (5 stages)              | Where does retrieval break? (Part 3 decomposition)      | κ 0.509 (Moderate)  |

The audit, not the model, set the ranking

1. Structure, not sentiment: Every review was converted into a structured, typed record with fixed vocabularies, not qualitative impression tags.

2. Pre-registered rules: Hypotheses H1–H5 and decision thresholds were set before extraction, so the data could not be read to fit them.

3. The audit overruled the ranking: When the blind audit showed hypothesis agreement was low (Jaccard 0.347) while failure stage agreement was solid (κ 0.509), the engine demoted hypothesis ranking and pivoted to failure stages.

[retrieval-discovery-engine.vercel.app](https://retrieval-discovery-engine.vercel.app)

# People keep "roughly when" and lose the exact date, the one thing search needs

Engine and survey agree: the time is kept, the date is lost

| Question      | Engine (144 Specific Attempts)                                           | Survey (n=15 / 14 Dedup)                                          |
|---------------|--------------------------------------------------------------------------|-------------------------------------------------------------------|
| Photo Kinds   | Personal photos 80 · multi-photo 26 · videos 16 · screenshots 7 · docs 3 | All 3 real-trouble cases were document or medicine photos         |
| Remembered    | Roughly when 40 · object 17 · exact date 14 · text 12 · people 10        | Roughly when 8 · object 7 · people with me 6                      |
| Forgotten     | The date 37 · album 29 · exact words 10 · place 10                       | When it was taken 9 · words to search 9                           |
| Search Method | 20 of 32 queries are 1 word ("dog"); 10 name a time ("Halloween 2024")   | First move: scroll 9/15; own-app: 4 of 5 strip time to bare nouns |

O1 is the largest group with a memory search ignores

O1 (Rough time or an event): 42 → 37 after exclusions (18 never surfaced, 12 misread).

O8 (No clue at all): 47 attempts, larger, but there is no memory for search to use. O1 is the largest group where memory exists and search ignores it.

Excluded: O7 exact date known (14) · O9 path moved by an update (12).

Episodic time wins; dead-end recovery loses

H1 Episodic time: SUPPORTED (root cause).

Roughly when is the #1 kept cue (40); the date is the #1 lost cue (37).

H2 Recognition: SUPPORTED (secondary). 41.7% never surfaced; 7 of 15 ended unsure.

H3 Dead-end recovery: REFINED, NOT THE LEAD. Overruled by the audit: recovery is 0.7% of failures (stage κ 0.509).

H4 Hinglish: NOT TESTED. Weak signal (2 of 15 survey, 1 of 6 tests); kept as a design choice.

H5 Text not indexed: WEAKLY SUPPORTED (3 docs; the index scores 0.000 on text in images).

H6 Path changed: PRESENT BUT MINOR (8.3%), added post hoc.

77% of the 144 failures break at I or S, before a retry

|                  |            |
|------------------|------------|
| S Never surfaced | 60 (41.7%) |
| I Clue misread   | 51 (35.4%) |
| E · R · retry    | 2 · 2 · 1  |

# Most people scroll before they search, and nearly half end close but unsure

**A structured survey gave 15 stories we could score**

Structured self-serve questionnaire (n=15, 23–30 Sep): every answer maps onto the engine's vocabulary.

**Why:** 100% scoreable stories (vs 8.6% of public posts); reached 15 people vs 2 call contacts.

**Costs:** self-reported, convenience sample; 8 of 15 fit the segment.

[Survey questions ↗](#)

| The survey repeats the engine's pattern |                   |                  |
|-----------------------------------------|-------------------|------------------|
| Metric                                  | Engine (144)      | Survey (15 / 14) |
| Most remembered                         | Roughly when (40) | Roughly when (8) |
| Ended unsure                            | Not measurable    | 7 of 15 (47%)    |
| First move: scroll                      | 15 workarounds    | 9 of 15 (60%)    |

**It costs most people over five minutes**

Trips and outings, a photo for LinkedIn, a medical report.

**12 of 15 spent 5+ minutes**, past the brief's five-minute bar; 3 had to ask someone or get the document again.

| 4 of 5 kept roughly when; only 1 put it in the search |                |                             |                                      |                |             |
|-------------------------------------------------------|----------------|-----------------------------|--------------------------------------|----------------|-------------|
| ID                                                    | Target Photo   | What They Remembered        | What They Typed                      | Break          | Outcome     |
| S04                                                   | Wedding photo  | Roughly when, event, people | "wedding, pichle saal diwali"        | Interpretation | Unsure      |
| S03                                                   | Personal photo | Roughly when, event, object | "gym"                                | Recognition    | Unsure      |
| S08                                                   | Personal photo | Roughly when, event         | (Didn't know what to type; scrolled) | Expression     | Found       |
| S12                                                   | Personal photo | Roughly when, event         | (Never typed; opened album)          | Expression     | Unsure      |
| S06                                                   | Medicine bill  | Object, text in bill        | "medicine, bill"                     | Surfacing      | Never found |

"wedding, pichle saal diwali" — S04 (Hinglish: a festival and relative year, not a calendar date)

"medicine, bill" — S06 (Never found; had to get the document again)

**Ask Photos:** 7 of 15 had never heard of it; all 4 who used it got "related photos, but not the one".

# Our segment can place the moment only roughly, and search can't turn "roughly" into dates

Defined by what they remember, not by age  
**Long-time Google Photos users searching for a personal moment they can place only roughly.**

Defined by **cognitive memory state** (what they remember), not demographics: "around Diwali", "my sister's graduation", "last winter".

| Segment Sizing Filter                    | Engine (144)    | Survey (15)    |
|------------------------------------------|-----------------|----------------|
| Kept rough time or event (O1)            | 42 (29%)        | 8 (53%)        |
| <b>Target Segment (After Exclusions)</b> | <b>37 (26%)</b> | <b>8 (53%)</b> |
| Personal photo or video                  | 30 of 37        | All 8          |
| Large library (5,000+ items)             | Not measured    | 5 of 8         |

**42 → 37:** minus 2 who also knew the exact date (plain search works) and 3 whose browse path moved in an update (H6). Sync and corruption issues are out of scope.

Memory stores episodes; Photos stores items and dates



"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

- Asking why, for the 37 segment attempts:
- 1. **Why isn't the photo found?** It never surfaces (18 of 37).
  - 2. **Why not?** Search misreads the clue (12): "last Diwali" is neither a keyword nor a date.
  - 3. **Why does that matter?** People keep the event and lose the date (the #1 lost cue, 37 of 144).
  - 4. **Root cause:** nothing turns "roughly when" into a date window.  
*Real app: 4 of 5 testers typed bare nouns ("vacation", "gym"); 0 of 5 found the photo.*

# People already scroll to "roughly when" by hand and still end unsure, so the product should take that step for them

## What users can't do, and why

"When people want a photo they can place only roughly ("last Diwali"), they can describe the moment but can't make Photos show it: search needs a date or keyword they've forgotten. So they scroll by hand, and nearly half end unsure."

## The Workaround is the Design Blueprint

**Engine:** 15 of 21 workarounds are manual scrolling (7 of 10 failed): *"Every time I scrolled to the right time frame, it kept loading more pictures..."*

**Survey:** Scrolling was the first move for 9 of 15. 5 of those 9 still ended unsure.

**Insight:** people already retrieve by moment, by hand.  
**The product should find the time window and group the moments.**

## Google already holds the data this needs

| Option                | How you search                                         |
|-----------------------|--------------------------------------------------------|
| Apple Photos (iOS 18) | Describe what's in the photo                           |
| Ask Photos            | Ask Gemini; all 7 who tried got "related, not the one" |
| Memory Trails         | <b>Roughly when → date window → moments, with why</b>  |

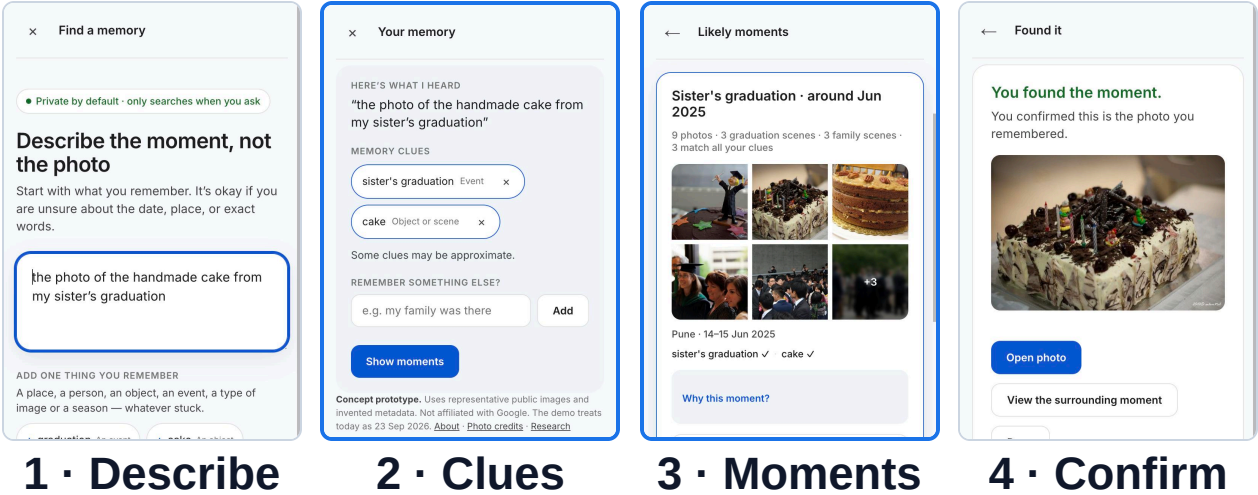
**Why Google can win:** it already holds every user's dates, places and faces, and could read events from Calendar. No standalone app has that data.

## Five times the evidence changed the plan

|                                                            |                                                               |                                                                 |                                                                    |                                                               |
|------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
| <b>1. Metric:</b> per-search success → <b>per-user URR</b> | <b>2. Outcomes:</b> recovery agent → <b>5 stages + browse</b> | <b>3. Discovery:</b> H3 ranked top → <b>overturned by audit</b> | <b>4. Reported behaviour:</b> retry search → <b>scroll by hand</b> | <b>5. Problem:</b> "search is bad" → <b>items vs. moments</b> |
|------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|

# Memory Trails turns "last Diwali" into a date window and shows moments, not a grid

**It sits inside search, where the failure happens**  
**Where it lives:** inside Photos search. When a search fails, "Can't describe it?" opens it with the words carried over.



[memory-trails-v2.vercel.app](https://memory-trails-v2.vercel.app) Blue frame = AI step

- Four mechanisms behind the gain**
- 1. **Clues** → **date window**: "pichle saal Diwali"
  - 2. **Moments, not a grid**: "3 match all your clues"
  - 3. **"Why this moment?"**: matched vs approximate
  - 4. **Soft scoring**: a wrong date lowers rank, never hides

| Big gain with rich memory, modest with one vague cue |                                                                |            |                |                |
|------------------------------------------------------|----------------------------------------------------------------|------------|----------------|----------------|
| Cue Dropout Level                                    | What the Query Keeps                                           | Plain CLIP | Memory Trails  | Moment@5       |
| L3 (Fully described)                                 | Vague time + exact place + library word                        | 0.172      | 0.962          | 0.933          |
| L2 (Two vague cues)                                  | Paraphrased content + 2 vague cues                             | 0.209      | 0.276          | 0.333          |
| L1 (One vague cue)                                   | Paraphrased content + 1 vague time cue                         | 0.242      | 0.309          | 0.333          |
| L0 (Content only)                                    | Pure scene description, no metadata                            | 0.312      | 0.312          | 0.267          |
| Weighted by real memory                              | Each level weighted by how often the 144 real attempts keep it | 0.259      | 0.334 (+0.075) | 0.336 (+0.111) |

**Read this first:** L3's 0.962 needs three cues; only 4% (6 of 144) of real attempts kept that many. Weighted by real memory, recall gains **+0.075** and moment@5 **+0.111**: grouping into moments helps more than flat ranking.

**Scope:** tests the re-entry interaction on a 1,282-photo Creative Commons library, not production-scale accuracy.

# 5 of 6 testers were sure they found the photo, but "year before last" still breaks it

| ID  | Own Photos search → result           | Task 1 (cat)      | Task 2 (dog)      | Sure | vs GP | Would use  |
|-----|--------------------------------------|-------------------|-------------------|------|-------|------------|
| R01 | "Wedding before COVID" → wrong years | Found, sure       | Not found         | 5/5  | 5/5   | Every time |
| R02 | "gym" → can't tell which             | Found, sure       | Found (1-tap alt) | 4/5  | 4/5   | When fails |
| R03 | (Apple Photos user)                  | Found, sure       | Found (nearby)    | 5/5  | —     | Every time |
| R04 | "vacation" → wrong years             | Found, sure       | Found (1-tap alt) | 4/5  | 4/5   | When fails |
| R05 | "Wedding event" → can't tell which   | Found, sure       | Found (clue edit) | 4/5  | 4/5   | Every time |
| R06 | "Kerala photos" → can't tell which   | Right day, unsure | Found (1-tap alt) | 4/5  | 3/5   | Every time |

6 of 6  
Fit the segment

4.33 / 5  
Task 1 sureness

4.0 / 5  
Ease vs Photos

0 of 5  
Found it in own Photos

MVP test questions ↗

memory-trails-v2.vercel.app ↗

## Testers valued the moment, not a grid

- "Shows the moment, not a grid" — R02
- "Moving to nearby months, like scrolling but faster" — R03
- "Works without knowing what to type" — R04
- "Finds exact month & year... understands mix of Hindi and English" — R06

## Three fixes for the next version

- Compound time:** R01 stuck on "pichle ke pichle saal" (year before last). *Fix:* parse compound offsets; editable year.
- Clues missed:** R04 didn't notice them. *Fix:* high-contrast "Showing Diwali 2025 · Tap to change".
- Speed:** R05, R06 wanted faster loading. *Fix:* pre-cache moments.

# Success is URR rising from 40% to 45% in a user-level A/B test, with an early signal per step

**One number: users who reach the photo in 28 days**  
**Definition:** share of users with a vague-memory search in 28 days who reach the photo. Counts users, not searches.  
**Baseline 40% → target 45%.** The baseline is a self-reported proxy (6 of 15 sure they found it). +5 pts is the smallest lift worth shipping.  
**Test:** 8 weeks, randomised by user; ~1,600 users per arm for 80% power at  $\alpha$  0.05.

**Recall and confidence already moved on the MVP**  
Recall@20 weighted by real memory **0.259 → 0.334** · moment@5 **+0.111** · tester sureness **4.33 / 5** · ease vs Photos **4.0 / 5**

**Ship, extend or stop is decided in advance**  
**Ship** if URR rises  $\geq 5$  pts and no guardrail trips. **Extend** if the lift is positive but under 5. **Stop** if false confirmation passes 3%.

Each URR term gets an early signal

| Metric                           | Term           | Target      |
|----------------------------------|----------------|-------------|
| Entry rate after a failed search | Expression     | $\geq 12\%$ |
| Clue correction (chip edited)    | Interpretation | 15–25%      |
| "Not in any of these?" rate      | Surfacing      | Falling     |
| Confirm after opening a moment   | Recognition    | $\geq 55\%$ |

A wrong 'found it' stops the launch

|                                                     |                    |
|-----------------------------------------------------|--------------------|
| False confirmation (confirmed, then kept searching) | <b>&lt;3%</b>      |
| P95 time to first moment                            | <b>&lt;1.5 s</b>   |
| Abandonment before the first moment                 | No rise vs control |

Diagnostics explain any movement

Recall@20 by cue type · near-miss recovery rate · "Why this moment?" views before confirming

# Biggest risk: the gain is largest where memory is richest, and the evidence is still small

Each risk has a mitigation built or planned

| Risk                                                                       | Mitigation & evidence                                                                  |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>R1. The win sits where memory is richest</b> (+0.79 at L3, +0.07 at L1) | Lean on moment recognition (moment@5 0.333 at L1–L2); report L1 separately in the A/B. |
| <b>R2. Date filters hide photos</b>                                        | <b>Soft scoring built:</b> L3 recall 0.851 → 0.962.                                    |
| <b>R3. False confirmation</b> (worst failure)                              | Never auto-confirm; show the surrounding photos; stop launch above 3%.                 |
| <b>R4. Can't read text in photos</b> (0.000 on receipts)                   | Add OCR text to the index.                                                             |
| <b>R5. Ask Photos may already suffice</b>                                  | 4 of 4 survey and 3 of 3 test users got "related photos, not the one".                 |
| <b>R6. Sensitive memories</b> (medical, relationships)                     | User-initiated only; the session persists nothing.                                     |

## The evidence is small and mostly self-reported

- Play Store is 86.2% of engine posts
- Survey n = 15 vs a target of 30, self-reported
- MVP tests n = 6, unmoderated, no session logs
- URR baselines modelled; no telemetry
- Eval library: 1,282 CC photos, synthetic episodes
- Ranking weights are judgement
- Hinglish (H4) is a weak signal only

## Next: compound dates, clearer clues, a one-market test

1. Parse compound dates ("pichle ke pichle saal")
2. Make the active date clue unmissable
3. Run the A/B in one market